

A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions:

Previous Normal Boxes and Box Relations

This note accompanies the GitHub repository [1] of *A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions* [2]. The handwritten proofs and the soundness checks in the repository use a different construction of the normal boxes A , B , and D from the one in the paper. We record this construction here, together with the action of each normal box on the Pauli generators and the resulting 66 box relations BR1 to BR66, so that the repository is self-contained.

With this construction, BR1 to BR66 form a complete set of box relations. The repository then follows the same steps as the paper. It reduces these box relations to symplectic relations and shows that, once lifted with Pauli corrections, they follow from the rules in Figure 1 of [2]. Therefore, this repository also gives a complete proof of the main result of the paper. The only difference is that the more involved normal boxes recorded here lead to 66 box relations instead of 42.

1 Scope

Below, we summarise the scope of this note and the parts of the repository [1] that rely on the previous normal boxes, as specified in Section 3.

- The soundness checks of BR1 to BR66 in `notebooks/BoxRelation-Soundness.ipynb`;
- The handwritten derivations of the box relations in `box-relation-derivations`;
- The handwritten relation reduction in `box-relation-reduction`;
- The file `notebooks/Python-verification-convention.pdf`, which records the symplectic conventions and sanity checks.

The box relations are numbered as in `BoxRelation-Soundness.ipynb`. This numbering is internal to the repository and differs from the order of the 42 box relations in Appendix F of the paper [2].

2 Conventions

We use the same notation as in the paper [2]. Throughout the note, p is an odd prime, $\omega = e^{2\pi i/p}$, and $\mathbb{Z}_p^*$ is the multiplicative group of non-zero elements of $\mathbb{Z}_p$. Exponents of ω and of gates of order p are read in $\mathbb{Z}_p$. For example, $S^{a^{-1}}$ is an integer power of S .

3 The Previous Normal Boxes

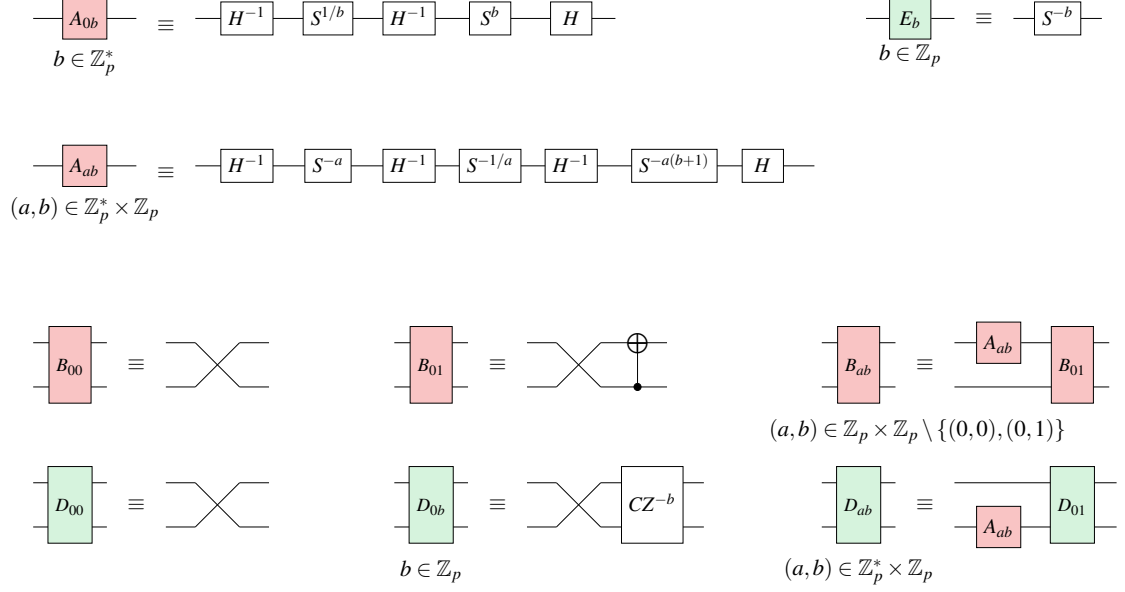
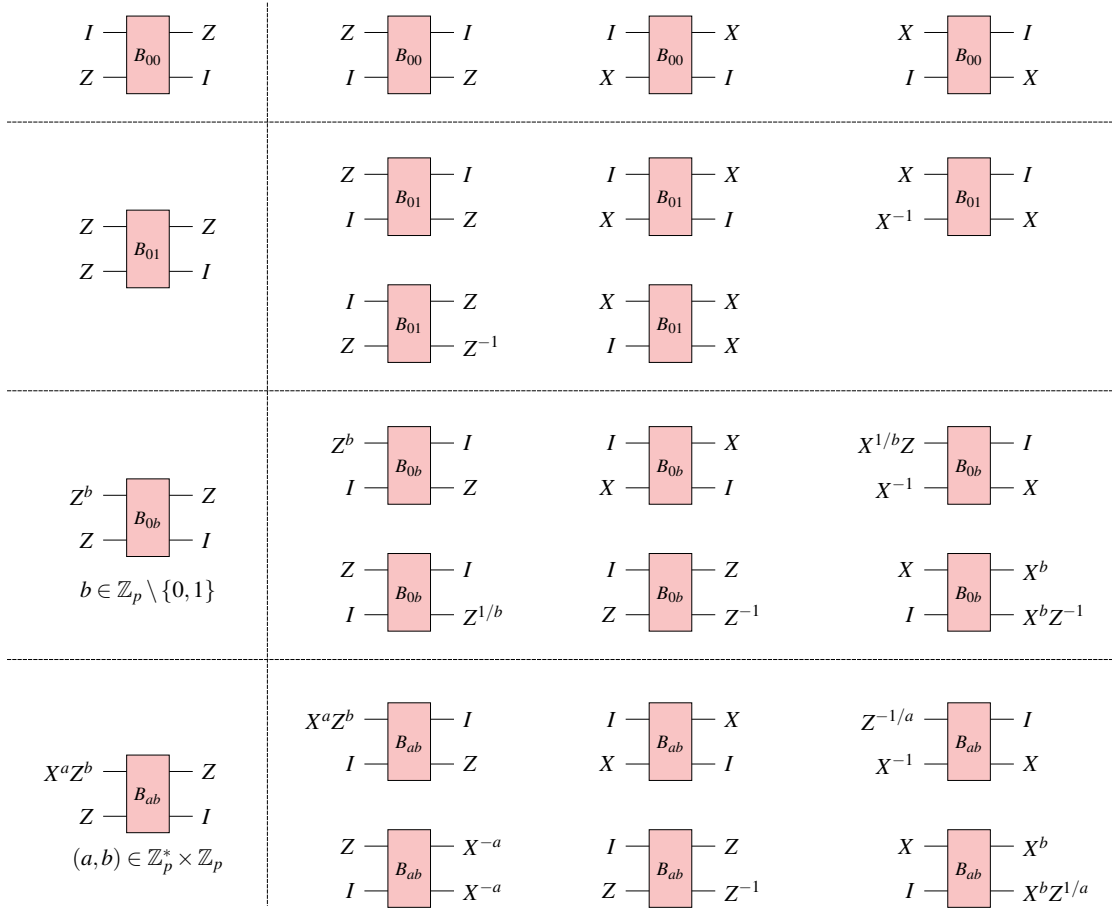


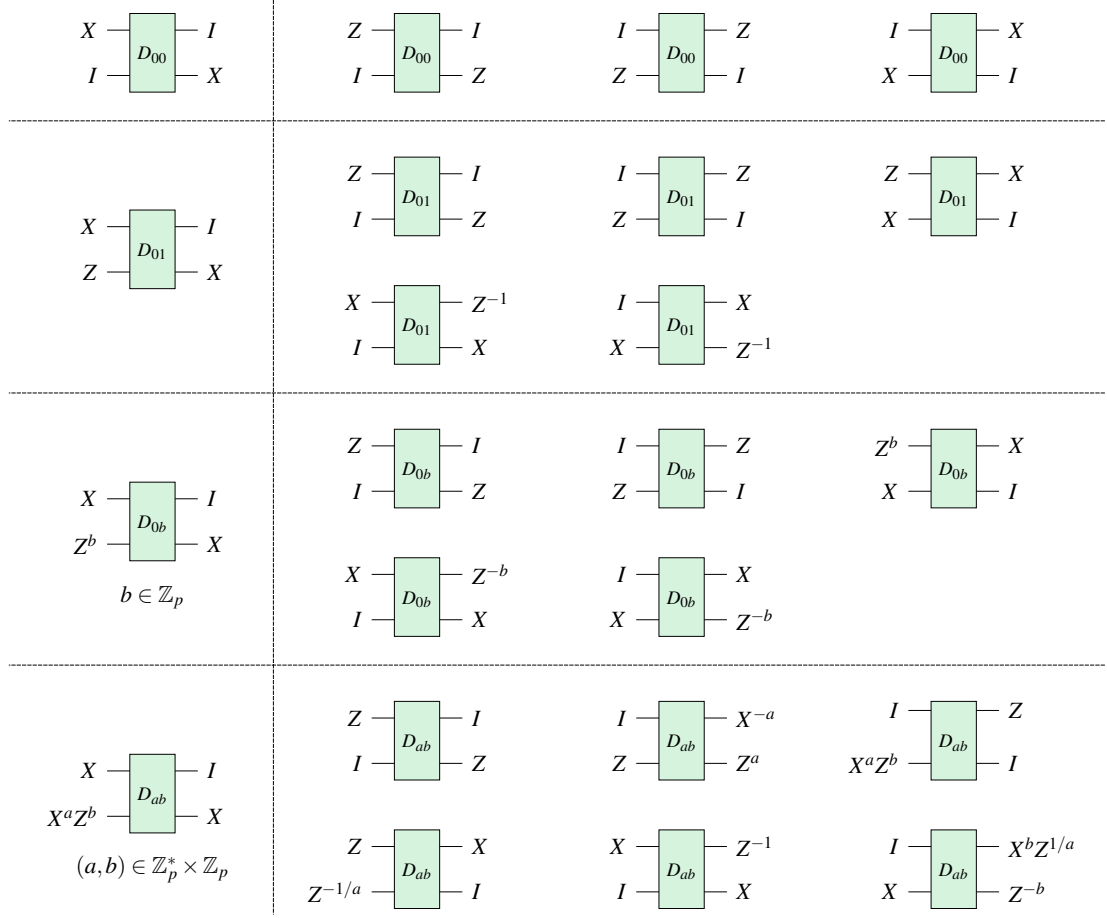
Figure 1: The previous construction of the normal boxes.

4 Action of the Normal Boxes on Pauli Generators

$$\begin{array}{ccc}
 \frac{Z^b}{X^{1/b}Z} \boxed{A_{0b}} \frac{Z}{X} & \frac{Z}{X} \boxed{A_{0b}} \frac{Z^{1/b}}{X^b Z^{-1}} & b \in \mathbb{Z}_p^* \\
 \\
 \frac{X^a Z^b}{Z^{-1/a}} \boxed{A_{ab}} \frac{Z}{X} & \frac{Z}{X} \boxed{A_{ab}} \frac{X^{-a}}{X^b Z^{1/a}} & (a,b) \in \mathbb{Z}_p^* \times \mathbb{Z}_p
 \end{array}$$

Figure 2: The action of A boxes on the Pauli generators up to phase.

Figure 3: The action of B boxes on the Pauli generators up to phase.

Figure 4: The action of D boxes on the Pauli generators up to phase.

$$\frac{Z}{X Z^b} \begin{array}{|c|} \hline E_b \\ \hline \end{array} \frac{Z}{X} \quad \frac{Z}{X} \begin{array}{|c|} \hline E_b \\ \hline \end{array} \frac{Z}{X Z^{-b}} \quad , \quad b \in \mathbb{Z}_p$$

Figure 5: The action of E boxes on the Pauli generators up to phase.

5 Box Relations

Each box relation pushes a generator H , S , or CZ through one or two normal boxes in Figure 1. We group them by the number of qudits they act on.

5.1 Single-Qudit Box Relations

$$\begin{aligned}
 & \text{---} [H] \text{---} [A_{0b}] \text{---} \equiv_s \text{---} [A_{b0}] \text{---} [S^{-1/b}] \text{---} & b \in \mathbb{Z}_p^* \\
 & \text{---} [H] \text{---} [A_{a0}] \text{---} \equiv_s \text{---} [A_{0,-a}] \text{---} [S^{-1/a}] \text{---} & a \in \mathbb{Z}_p^* \\
 & \text{---} [H] \text{---} [A_{ab}] \text{---} \equiv_s \text{---} [A_{b,-a}] \text{---} [S^{1/ab}] \text{---} & a, b \in \mathbb{Z}_p^*
 \end{aligned}$$

Figure 6: Pushing H through an A box, up to Pauli correction.

$$\begin{aligned}
 & \text{---} [S] \text{---} [A_{0b}] \text{---} \equiv_s \text{---} [A_{0b}] \text{---} [S^{1/b^2}] \text{---} & b \in \mathbb{Z}_p^* \\
 & \text{---} [S] \text{---} [A_{ab}] \text{---} \equiv_s \text{---} [A_{a,b-a}] \text{---} & (a, b) \in \mathbb{Z}_p^* \times \mathbb{Z}_p
 \end{aligned}$$

Figure 7: Pushing S through an A box, up to Pauli correction.

$$\text{---} [S] \text{---} [E_b] \text{---} \equiv_s \text{---} [E_{b-1}] \text{---} \quad b \in \mathbb{Z}_p$$

Figure 8: Pushing S through an E box.

5.2 Two-Qudit Box Relations

$$\begin{aligned}
 & \text{---} \bullet \text{---} [A_{0b}] \text{---} \equiv_s \text{---} [A_{0b}] \text{---} [CZ^{1/b}] \text{---} & b \in \mathbb{Z}_p^* \\
 & \text{---} \bullet \text{---} [A_{ab}] \text{---} \equiv_s \text{---} [A_{0,-a}] \text{---} [B_{ab}] \text{---} [H] \text{---} [S^a] \text{---} [H^2] \text{---} \bullet \text{---} [H] \text{---} [S^{1/a}] \text{---} [H^3] \text{---} & (a, b) \in \mathbb{Z}_p^* \times \mathbb{Z}_p
 \end{aligned}$$

Figure 9: Pushing CZ through an A box, up to Pauli correction.

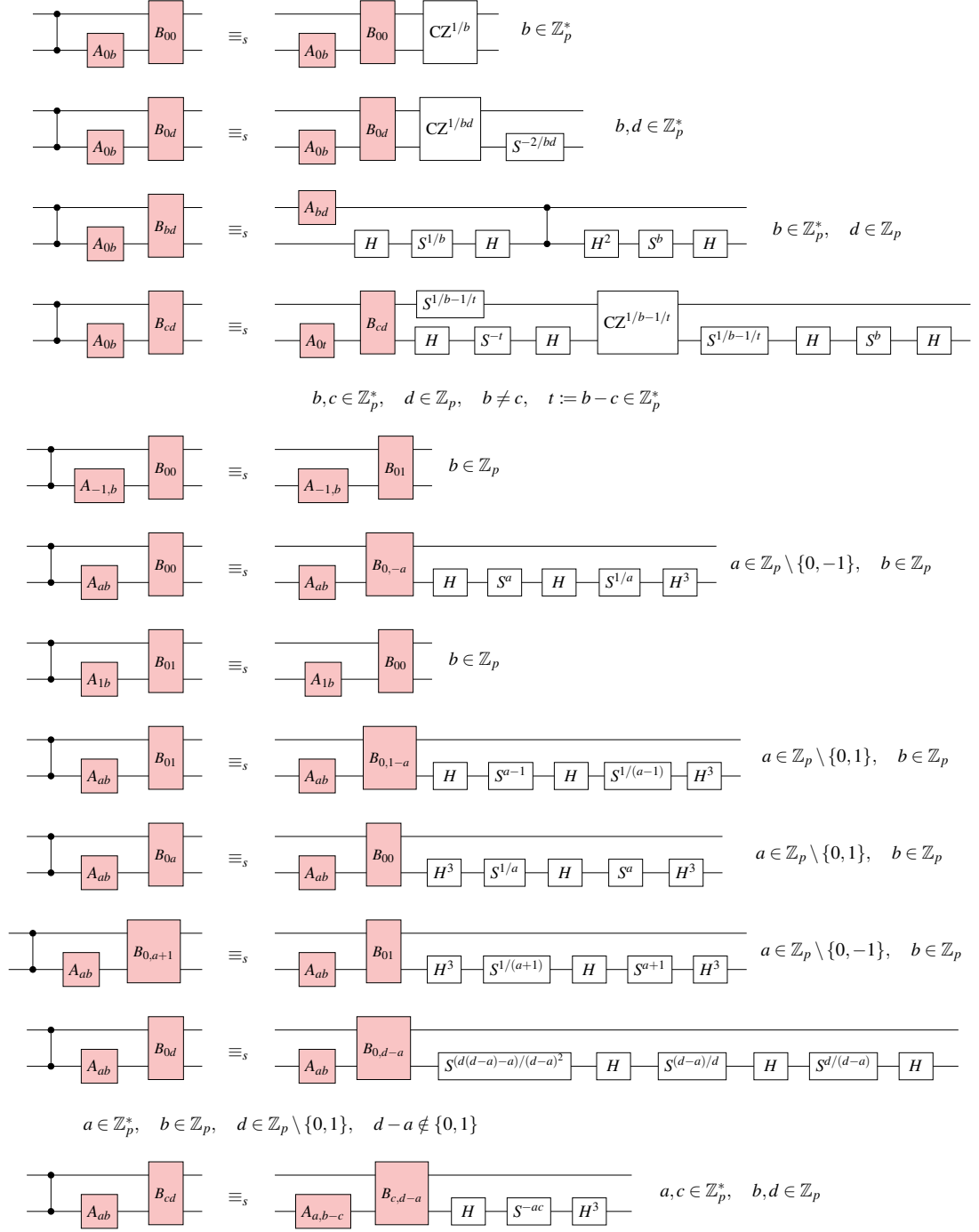


Figure 10: Pushing CZ through A and B boxes, up to Pauli correction.

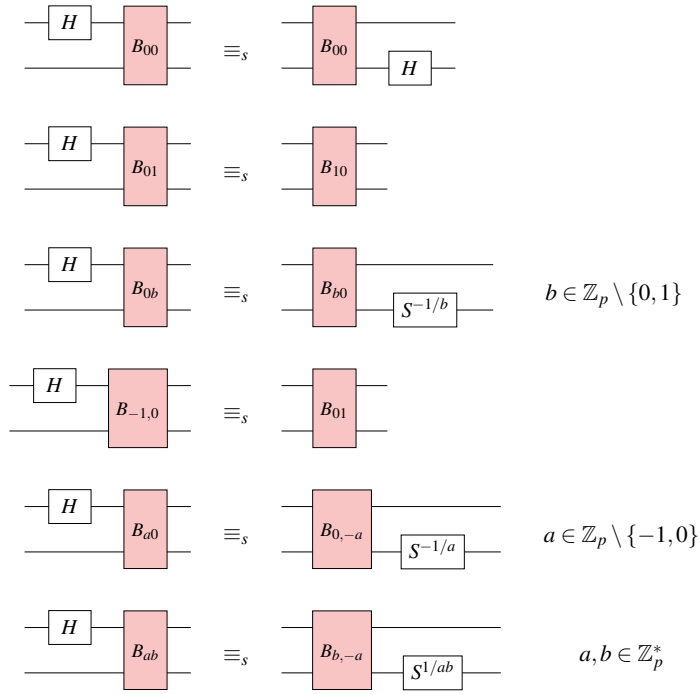


Figure 11: Pushing $H \otimes I$ through a B box, up to Pauli correction.

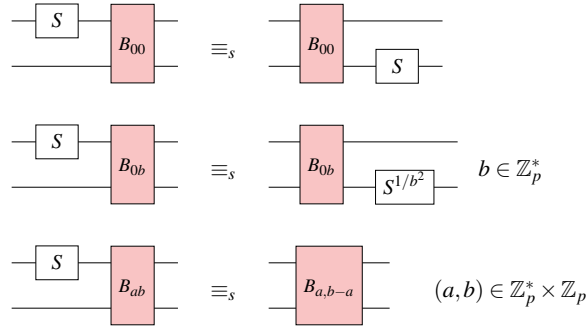
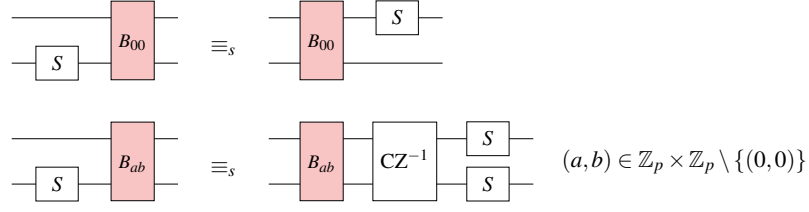
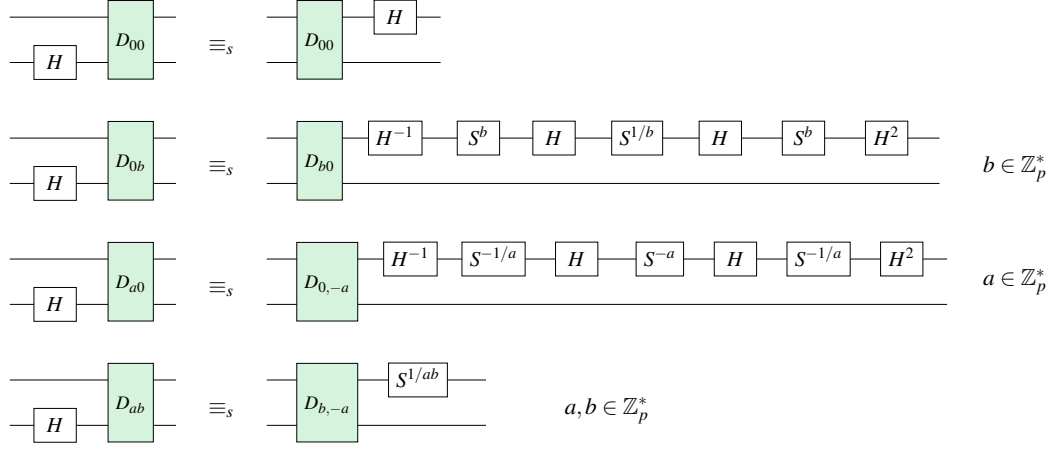
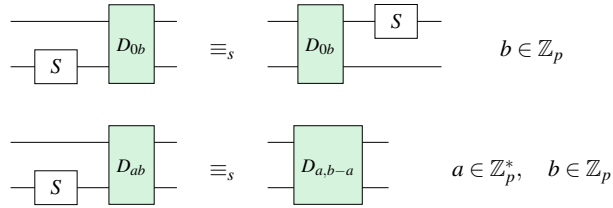
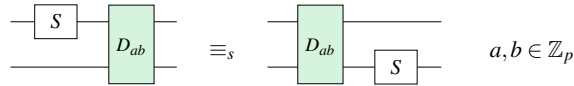


Figure 12: Pushing $S \otimes I$ through a B box, up to Pauli correction.

Figure 13: Pushing $I \otimes S$ through a B box, up to Pauli correction.Figure 14: Pushing $I \otimes H$ through a D box, up to Pauli correction.Figure 15: Pushing $I \otimes S$ through a D box, up to Pauli correction.Figure 16: Pushing $S \otimes I$ through a D box, up to Pauli correction.

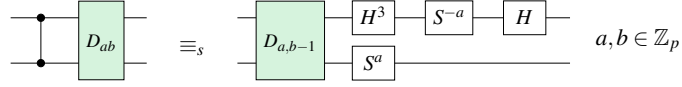


Figure 17: Pushing CZ through a D box, up to Pauli correction.

5.3 Three-Qudit Box Relations

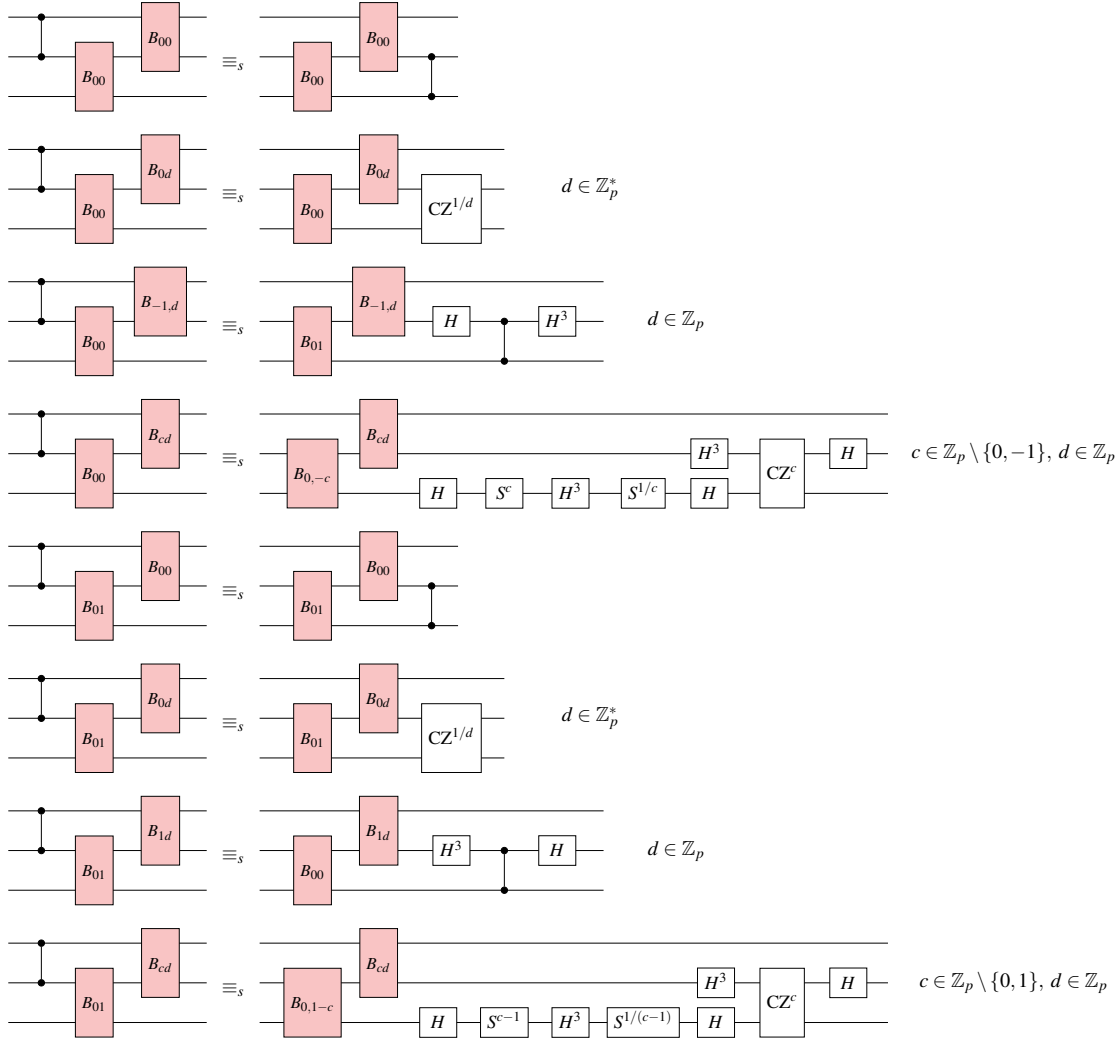


Figure 18: Pushing $\text{CZ} \otimes I$ through two B boxes— B_{00} or B_{01} followed by another B box, up to Pauli correction.

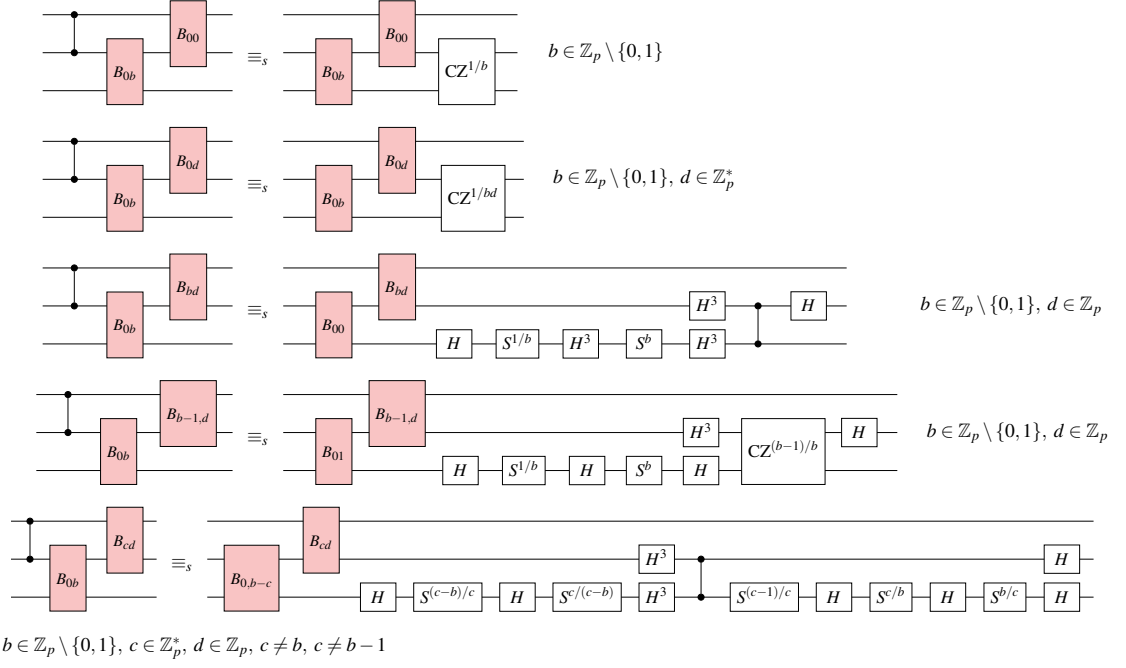


Figure 19: Pushing $\text{CZ} \otimes I$ through two B boxes— B_{0b} followed by another B box, up to Pauli correction. Here, $b \in \mathbb{Z}_p \setminus \{0, 1\}$.

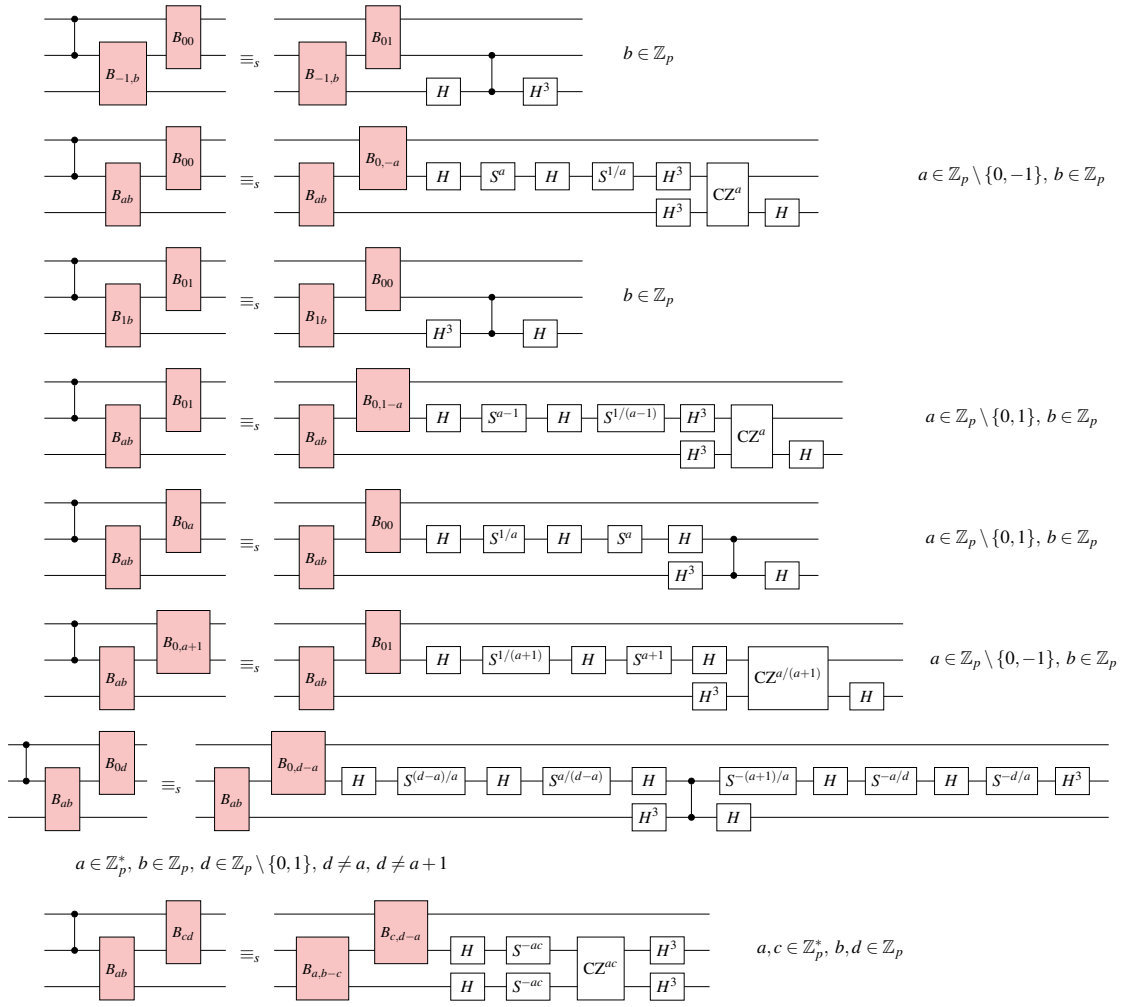
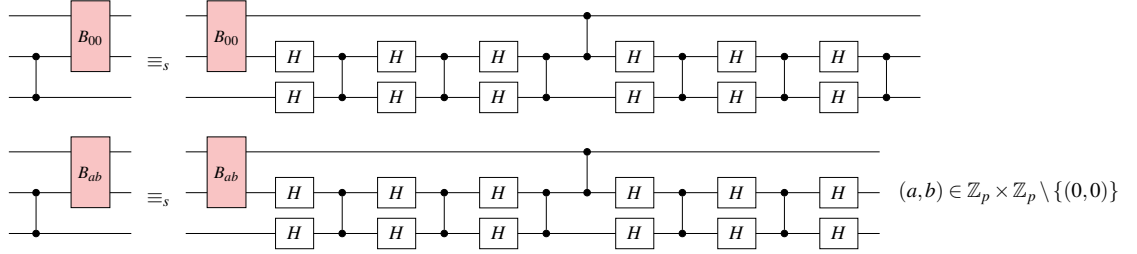
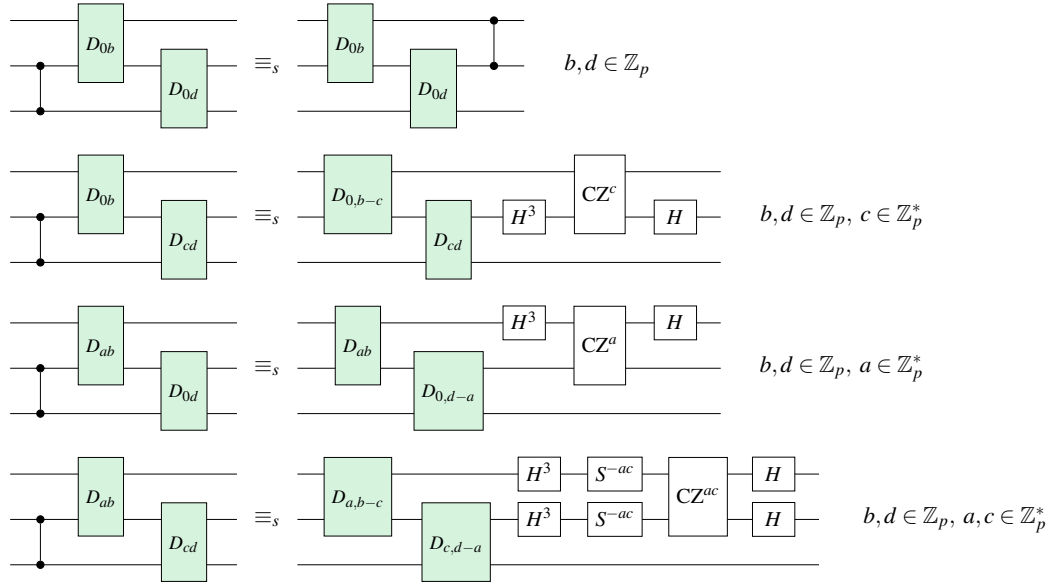


Figure 20: Pushing $\text{CZ} \otimes I$ through two B boxes— B_{ab} followed by another B box, up to Pauli correction. Here, $a \in \mathbb{Z}_p^*$ and $b \in \mathbb{Z}_p$.

Figure 21: Pushing $I \otimes CZ$ through a B box, up to Pauli correction.Figure 22: Pushing $I \otimes CZ$ through two D boxes, up to Pauli correction.

References

- [1] Xiaoning Bian, Sarah Meng Li, Neil J. Ross, John van de Wetering, and Yuming Zhao. A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions. <https://github.com/SarahMMMLi/QuditClifford>, 2026. Handwritten proofs of relation reduction.
- [2] Xiaoning Bian, Sarah Meng Li, Neil J. Ross, John van de Wetering, and Yuming Zhao. A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions. *arXiv preprint arXiv:2609.40106*, 2026.